A DRAUGHT OF THE EASTERN LAKES

The Canoe Traveller's Network

The instructor's key / Ordinary tandem team / Double carries / Model assumptions, not observed travel times

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I. THREE PRE-DESIGNED JOURNEYS

A - Inner circuit / 13.25-27.25 h

Clearwater - West Pike - Pine - Little Caribou - Caribou - Clearwater

B - Eastern circuit / 19.5-40.75 h

Clearwater - West Pike - East Pike - John - Little John - McFarland
Pine - Little Caribou - Caribou - Clearwater

C - Rose out and back / 17.25-33.25 h

Clearwater - Mountain - Watap - Rove - Rose; return by the same chain

Totals are moving + handling hours, not days or daily targets. Rest, camps, queues, difficult weather and windbound waits add time. Both directions use the same broad scenario envelope; wind direction changes which end of the range is plausible.

II. HOW THE HOURS ARE RECKONED

Paddle: geographic water-path distance, plus 15-30% indirect travel; 20-35% on more winding paths. Divide by 4 km/h (favourable) or 2 km/h (adverse but travelable). These are teaching assumptions.

Carry: 2 loaded walks at 2-3 km/h, 1 return at 3-4 km/h, plus 10-20 minutes of handling per connection. One rod = 5.0292 m.

Bounds round outward to quarter hours. A 0-0.25 h label means a brief nonzero leg. All rod counts are derived, not official counts.

III. HIGH GROUND & UNCERTAINTY

H01 ~600 m / near Clearwater

H04 ~625 m / near Clearwater

H08 ~620 m / near West Pike

H10 ~600 m / near Pine

H12 ~600 m / near East Pike

These are bare-earth terrain candidates, not verified visible peaks. Tree cover can obscure them. Their schematic placement is illustrative; use the separate geographic project for true positions and bearings.

Carry a: USFS line is incomplete; published digitized GPX supplies an approximate full passage.
Carries h and i: published sketches, no matching USFS geometry. Water links f and m are not carries.

Clearwater-Deer was omitted: the recorded trail stops short of the lakes. Deer connects via Caribou.

Rose is a terminal branch. Its T point, like those on Deer, Alder and Crystal, is a hypothetical water turnaround, not a campsite or landing.

SOURCES & REPRODUCTION

MN DNR waters / USFS trails / MnGeo terrain
Friends of BWCA digitized loop / Parks Canada history
Timing context: Poudre Paddlers; NPS wind variability
Full citations and limitations: docs/SOURCES.md

The Canoe Traveller's Network

The reckoning of each passage / Source-backed distances; modeled hours / All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
a	Clearwater / West Pike		carry	1.04	-	1-1.75 h	approximate*
b	West Pike / Pine		carry	1.62	-	1.5-2.5 h	USFS + shore
c	West Pike / East Pike		carry	0.92	-	1-1.75 h	USFS + shore
d	East Pike / John		carry	1.07	-	1-2 h	USFS + shore
e	John / Little John		carry	0.08	-	0-0.5 h	USFS + shore
f	Little John / McFarland		river	0.03	-	0-0.25 h	approximate*
g	McFarland / Pine		carry	0.05	-	0-0.5 h	USFS + shore
h	Pine / Little Caribou		carry	0.40	-	0.5-1 h	approximate*
i	Little Caribou / Caribou		carry	0.10	-	0.25-0.5 h	approximate*
j	Caribou / Clearwater		carry	1.05	-	1-1.75 h	USFS + shore
k	Clearwater / Mountain		carry	0.42	-	0.5-1 h	USFS + shore
l	Mountain / Watap		carry	0.47	-	0.5-1 h	USFS + shore
n	Rove / Rose		carry	3.28	-	3-4.75 h	USFS + shore
o	Caribou / Deer		carry	0.24	-	0.25-0.75 h	USFS + shore
p	Pine / Canoe		carry	2.32	-	2.25-3.5 h	USFS + shore
q	Canoe / Alder		carry	0.13	-	0.25-0.75 h	USFS + shore
r	Canoe / Crystal		carry	0.26	-	0.25-0.75 h	USFS + shore
m	Watap / Rove		narrows	<0.01	-	0-0.25 h	approximate*
W01	Clearwater	S-a	paddle	9.82	15-30%	2.75-6.5 h	water model
W02	Clearwater	S-j	paddle	1.97	15-30%	0.5-1.5 h	water model
W03	Clearwater	S-k	paddle	7.54	15-30%	2-5 h	water model
W04	Clearwater	a-j	paddle	8.48	20-35%	2.5-5.75 h	water model
W05	Clearwater	a-k	paddle	2.54	15-30%	0.5-1.75 h	water model
W06	Clearwater	j-k	paddle	6.23	15-30%	1.75-4.25 h	water model
W07	West Pike	a-b	paddle	6.07	15-30%	1.5-4 h	water model
W08	West Pike	a-c	paddle	6.07	15-30%	1.5-4 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

Paddle points are identified by the connecting letters on the chart. S = departure; T = hypothetical turning place.

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The reckoning of each passage ⁄ Source-backed distances; modeled hours ⁄ All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
W09	West Pike	b-c	paddle	0.00	15-30%	same landing	water model
W10	East Pike	c-d	paddle	5.83	15-30%	1.5-4 h	water model
W11	John	d-e	paddle	2.09	15-30%	0.5-1.5 h	water model
W12	Little John	e-f	paddle	0.90	15-30%	0.25-0.75 h	water model
W13	McFarland	f-g	paddle	4.00	20-35%	1-2.75 h	water model
W14	Pine	b-g	paddle	5.49	15-30%	1.5-3.75 h	water model
W15	Pine	b-h	paddle	6.12	15-30%	1.75-4 h	water model
W16	Pine	b-p	paddle	6.21	15-30%	1.75-4.25 h	water model
W17	Pine	g-h	paddle	11.51	15-30%	3.25-7.5 h	water model
W18	Pine	g-p	paddle	11.61	15-30%	3.25-7.75 h	water model
W19	Pine	h-p	paddle	0.13	15-30%	0-0.25 h	water model
W20	Little Caribou	h-i	paddle	1.00	15-30%	0.25-0.75 h	water model
W21	Caribou	i-j	paddle	5.52	15-30%	1.5-3.75 h	water model
W22	Caribou	i-o	paddle	5.37	15-30%	1.5-3.5 h	water model
W23	Caribou	j-o	paddle	0.22	15-30%	0-0.25 h	water model
W24	Mountain	k-l	paddle	1.02	20-35%	0.25-0.75 h	water model
W25	Watap	l-m	paddle	3.26	15-30%	0.75-2.25 h	water model
W26	Rove	m-n	paddle	1.91	15-30%	0.5-1.25 h	water model
W27	Rose	T-n	paddle	1.57	15-30%	0.25-1.25 h	water model
W28	Deer	T-o	paddle	0.33	15-30%	0-0.25 h	water model
W29	Canoe	p-q	paddle	1.33	15-30%	0.25-1 h	water model
W30	Canoe	p-r	paddle	0.71	20-35%	0-0.5 h	water model
W31	Canoe	q-r	paddle	0.74	15-30%	0-0.5 h	water model
W32	Alder	T-q	paddle	1.14	15-30%	0.25-0.75 h	water model
W33	Crystal	T-r	paddle	1.82	15-30%	0.5-1.25 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

Paddle points are identified by the connecting letters on the chart. S = departure; T = hypothetical turning place.